

# Nithish Kumar R

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## ABOUT

AI & Data Science undergraduate focused on AI engineering, production ML systems, and LLM-powered applications using Python, FastAPI, Docker, and modern AI APIs. IEEE-published author at ICISS 2026.

## EDUCATION

**M Kumarasamy College of Engineering**  
*B.tech. Artificial Intelligence and Data Science*  
**Velammal Matric. Hr. Sec. School, Theni**  
*Computer Science and Mathematics*

Aug 2022 – Mar 2026  
CGPA: 7.25 (Scale: 10)  
June 2022  
Percentage: 70%

## SKILLS

**Languages:** Python, SQL  
**AI & ML :** LLMs, RAG Systems, CatBoost, Scikit-learn, NLP, Whisper STT  
**Backend & APIs:** FastAPI, Flask, REST APIs, Next.js, TypeScript  
**Databases:** PostgreSQL, pgvector, MySQL, Notion API  
**Deployment & Tools:** Docker, Google Cloud Run, Git, GitHub, Streamlit

## EXPERIENCE

**Database Administrator Intern**  
IMET Tooling India Pvt Ltd, Coimbatore

Jul 2024 – Mar 2026

- Designed and managed MySQL database schemas for a B2B bulk order management system, ensuring data integrity and performance
- Developed and optimized SQL queries and supported backend operations in a XAMPP-based environment
- Collaborated with cross-functional teams, contributing to system integration, testing, and debugging

## PUBLICATIONS

**AI-Driven and NLP-Powered Multilingual Tutor for Indian Languages**

Mar 4-6 2026

Vaishnavi J., Prasanth K. R., **Nithish Kumar R.**, Pragadheesh M., Hareesh K. | *8th International Conference on Intelligent Sustainable Systems (ICISS 2026)*, IEEE-affiliated | Paper ID: ICISS-0103 | [Accepted](#)

## PROJECTS

**Cognitive Memory Engine** | FastAPI, Next.js, PostgreSQL, pgvector, Gemini API

- Built and deployed a full-stack AI learning intelligence platform that turns study activity into structured memories, semantic retrieval, analytics, and personalized recommendations
- Implemented semantic memory using BAAI/bge-base-en-v1.5 embeddings with pgvector and HNSW indexing
- Built analytics and recommendation engines for study consistency, topic distribution, and neglected-area detection; deployed Dockerized FastAPI and Next.js services on Google Cloud Run with Neon PostgreSQL

**AI Personal Assistant System** | Python, Whisper, Ollama, Notion API

- Built a voice-enabled AI assistant integrating Whisper STT, LLM routing, and persistent memory workflows
- Integrated Notion APIs for contextual memory storage, diary generation, and retrieval-based workflows
- Developed modular Python pipelines for speech processing, memory retrieval, and LLM-powered responses

**Food Delivery ETA Prediction** | Python, CatBoost, Docker, Google Cloud Run

- Built and deployed an end-to-end ML pipeline for food delivery ETA prediction using production inference workflows
- Benchmarked CatBoost, XGBoost, Random Forest, and LightGBM models with CatBoost achieving best performance ( $R^2$ : 0.84)

## CERTIFICATION

**IBM AI Developer Professional Certificate** | IBM (Coursera)

Nov 2024

Covered generative AI, prompt engineering, Python, Flask, and AI application development.